

# Sai Tarun Sirapurapu

+1-937-329-2252 | [tarunsirapurapu@gmail.com](mailto:tarunsirapurapu@gmail.com) | [portfolio](#) | [in](#) [saitarunsirapurapu](#) | [tarungit98](#)

## OBJECTIVE

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Currently Developer at Reynolds and Reynolds with hands-on experience in React, TypeScript, Java and MySQL. Actively seeking roles in AI and machine learning where I can apply my professional development experience and academic background in deep learning, and open-source contributions to build intelligent, scalable, and user-focused applications.

## EXPERIENCE

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### • Reynolds & Reynolds

Aug 2025 - Present

Developer

Dayton, USA

- Migrated 3 legacy ColdFusion reports to React and TypeScript end-to-end, modernizing core dealership workflows while maintaining data integrity across production.
- Built dealer equity visualization and financial health dashboards in React and TypeScript used by 300+ automotive dealerships to identify profit opportunities and analyze/manage current portfolio in real-time.
- Engineered reporting interfaces displaying LTV ratios, equity positions, and payment trends across large dealer portfolios, giving dealerships a live view of at-risk accounts and refinancing opportunities.
- Owned SQL query optimization and API design for the analytics engine, improving report generation performance without degrading data accuracy across high-volume dealer datasets.
- Integrated React frontends with Node.js microservices and PostgreSQL using HeidiSQL to deliver full-stack features end-to-end across the platform.
- Incorporated AI-assisted code review into the dev workflow, cutting PR review time from ~2 days to 4–6 hours and review rounds from 1–2 to 1, with cleaner code style and fewer nitpick comments.
- Efficient use of Claude through skill.md, superpowers, brainstorming and all tools associated for cleaner throughput of work.

### • Dream Studio (Model Earth Project)

Feb 2025 - Aug 2025

Open-Source Contributor

Remote, USA

- Led development of the RealityStream model generation pipeline, dynamically sending geospatial and temporal data to six ML models (XGBoost, Random Forest, Neural Network MLP, etc.).
- Integrated global contributors' code into a unified Python app, improving system reliability and activating robust parameter handling from YAML configurations.
- Enhanced multi-model ML workflows with modular imports, conditional training logic, and reproducible pipelines.
- Implemented dynamic parameter parsing from YAML configurations to control training outputs and file saving behavior.
- Refactored model loading logic to conditionally import only selected ML models (e.g., XGBoost, Random Forest, SVM) based on config inputs.
- Assisted in developing performance aggregation and report generation pipelines using pandas and sklearn metrics.

### • CGI

Oct 2020 - Dec 2022

Software Engineer

Bangalore, India

- Built and maintained data-driven enterprise applications for telecom billing systems using Java Spring, SQL, and REST APIs.
- Conducted data analysis and algorithm refinement, achieving a 30% reduction in processing time for backend workflows.
- Applied security best practices and performed validation on SOAP/REST services using tools like Postman and SOAP UI.
- Integrated SQL procedures into backend microservices, applying object-oriented programming and structured data management.
- Led cloud-native service migration to AWS and Dockerized infrastructure to enable scalable deployments and analytics workloads.
- Participated in Agile development cycles, contributed to test automation, and documented decisions for reproducibility.

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### • University of Dayton

May 2023 - Dec 2024

- Assisted in graduate-level Deep Learning courses focused on neural networks, transformers, and generative models.
- Designed assignments on latent space visualization and implemented Variational Autoencoders (VAEs) to teach dimensionality reduction and representation learning.
- Conducted labs with Hugging Face models for NLP tasks including text classification and summarization.

- Supported grading and mentoring for student projects involving real-world datasets and model interpretability.
- Developed instructional material on ML fundamentals, emphasizing reproducibility, fairness, and evaluation metrics.
- Contributed to a dynamic academic environment, showcasing strong teamwork skills and reliability, and the capacity to address complex challenges in AI and robotics.

## EDUCATION

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- **University of Dayton** Jan 2023 - Dec 2024  
MS in Computer Science 3.83 GPA Dayton, USA
- **Keshav Memorial Institute of Technology** Aug 2016 - Sept 2020  
B.Tech in Electronics and Communication Engineering Hyderabad, India

## PROJECTS

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### • AI News Curator

Tools: Java, Spring Boot, React, Tailwind CSS, Docker, REST APIs, Maven



- Developed a full-stack personalized news aggregation system using NLP techniques for abstractive summarization, named entity recognition, and keyword extraction.
- Applied cosine similarity and topic modeling (LDA) to rank and cluster articles based on semantic relevance and user preferences.
- Designed scalable RESTful APIs to dynamically fetch, summarize, and rank content in near-real-time.
- Integrated pre-trained transformer-based models from Hugging Face (e.g., Microsoft MiniLM-L6-v2, BERT-base-uncased) for text summarization and language understanding.
- Engineered a modular backend architecture enabling continuous news ingestion and intelligent filtering.
- Containerized the backend using Docker for reproducibility and deployed to a scalable cloud infrastructure for production use.

### • Real Estate Analysis

Tools: Python, Flask, React, AWS, Nginx, Linear Regression, Pandas, Scikit-learn



- Reprocessed and cleaned Bangalore housing data, handling missing values and feature engineering.
- Implemented Linear Regression for price prediction and pickled the trained model for deployment, showcasing my skills in model development.
- Developed server and utility files in Python (Flask API) for backend processing emphasizing an event-driven architecture reflecting my OOP and MVC knowledge and deployed pickled model.
- Built a React-based frontend for user interaction and real-time predictions, optimizing for mobile responsiveness.
- Deployed the full-stack application on AWS using FileZilla and Nginx web server for hosting simulations, with a focus on persistence of data. Leveraged SQL Server for data integrity.

### • Real-Time Robotic Object Detection System

Tools: Robot Operating System 2, Python, Docker, OpenCV, You Only Look Once v8, Ubuntu, Hugging face and Deep java library.

- Developed a system integrating ROS2 and YOLO for detecting and classifying Mars rocks.
- Designed three ROS2 nodes for camera input, YOLO-based detection, and Turtlesim-based movement.
- Enabled real-time object processing with support for multi-camera inputs ensuring data readability.
- Coordinated robotic actions based on detection results in a simulated environment enhancing my understanding of runtime paradigms and event-driven programming.

## SKILLS

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### ◦ Programming Languages:

**Backend:** Java (Spring Boot), Python, Node JS

**Frontend:** JavaScript, React + TypeScript

**Database:** MySQL, Postgres SQL, Heidi SQL

### ◦ Deep Learning & Computer Vision

**Frameworks:** TensorFlow, Keras, Pytorch, Scikit-learn, OpenCV

**Models:** Neural Networks, Generative Adversarial Networks (GAN), Variable Auto Encoder (VAE), Denoising Diffusion Probabilistic Models (DDPM), Specialized in Single Shot Detector (SSD) MobileNet V2 optimization and deployment.

### ◦ Development Tools & Technologies

**Version Control:** Git, GitHub

**Containerization:** Docker

**Operating Systems:** Linux

**Data Visualization:** Tableau, MATLAB, Python libraries

**Cloud Platforms:** Experience with cloud deployment and microservices on AWS